

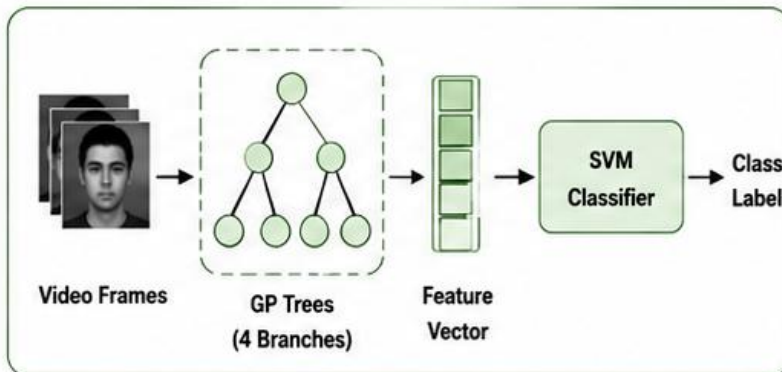
MARS – Week 7

Benchmarking Explainable Feature Extraction Methods for Facial Image Classification

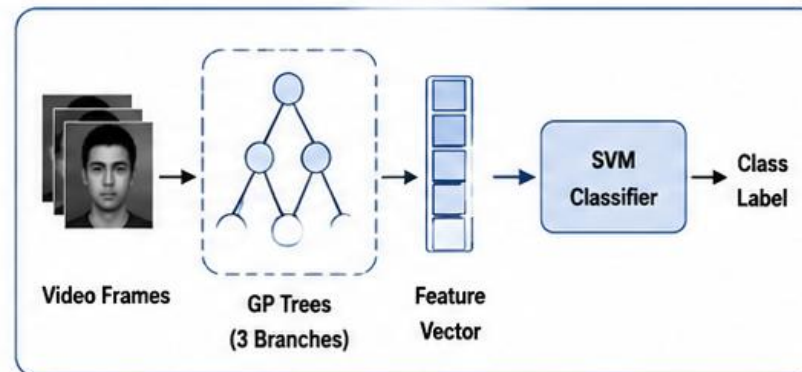
Comparing Handcrafted Features, Genetic Programming, Modified GP and Evolution of Heuristics (EOH)

James Davies

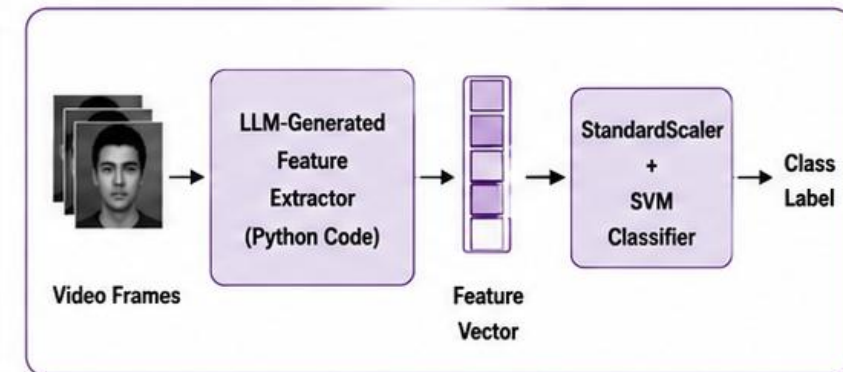
Original GP



Modified GP



LLM (EOH)



3 August 2026



Research Question

Can evolutionary algorithms and Large Language Models automatically discover compact, explainable feature representations that outperform handcrafted feature extraction for facial image classification?

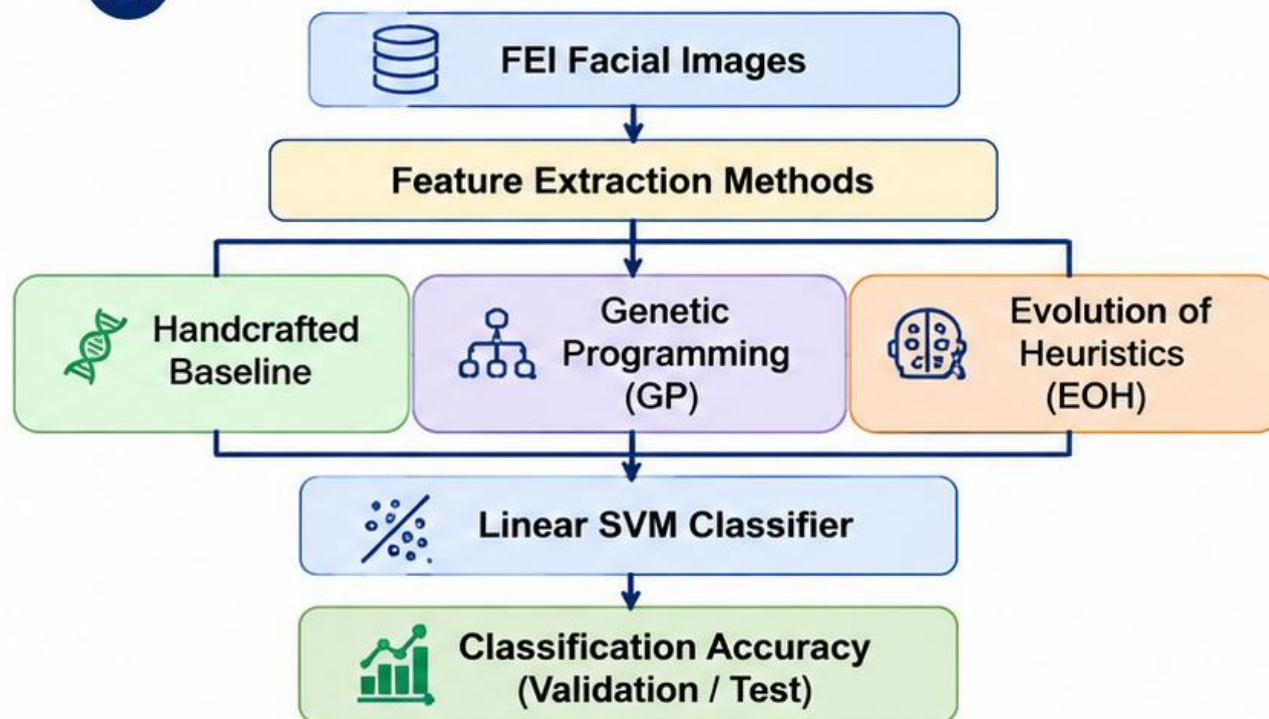


Research Objectives

- ✓ Develop an explainable **handcrafted baseline** for facial image classification.
- ✓ Compare Genetic Programming (**GP**), Modified Genetic Programming (**MGP**) and Evolution of Heuristics (**EOH**) under identical conditions.
- ✓ Investigate whether learned feature extractors provide improved performance while maintaining **interpretability**.
- ✓ Establish a **reproducible benchmarking framework** for fair and statistically meaningful comparisons.



Research Workflow



The project aims to discover compact, high-performing and explainable feature representations through evolutionary search and LLM-guided program synthesis.

Recap of Previous Week



Last week, the focus was on implementing the core methods and establishing the initial benchmarking pipeline.



WHAT WAS COMPLETED



Original GP Implemented

First working implementation of the original GP feature extraction framework.



Modified GP Implemented

Implemented the modified GP variant with changes to genetic operators and selection.



Initial EOH Framework Integrated

Integrated the Evolution of Heuristics (EOH) framework with DeepSeek API.



Initial Benchmarking Pipeline Developed

Developed the first version of the benchmarking pipeline for evaluating different methods.



SUPERVISOR FEEDBACK



Develop an explainable baseline

Create a handcrafted baseline using interpretable features as a reference point.



Control feature dimensionality

Ensure the same number of features across methods to enable fair comparison.



Build a common benchmark

Establish a unified evaluation framework with shared data splits and metrics.



Focus on robustness and explainability

Investigate generalisation, robustness and the interpretability of discovered features.



Analyse performance differences

Investigate why modified GP may behave differently from the original GP.



KEY MESSAGE

We have now implemented the core methods and received clear direction on how to build a rigorous, fair and explainable benchmarking framework.

4 Progress This Week



This week has focused on completing the benchmarking framework, establishing an explainable baseline, integrating all methods into the common pipeline, and obtaining the first EOH results.



MAJOR ACHIEVEMENTS THIS WEEK

1



Handcrafted Baseline Completed

Designed and implemented 8 interpretable features with full evaluation in the common pipeline.

2



Unified Benchmark Notebook Developed

Built a reproducible benchmarking framework with common preprocessing, data splits, classifier and metrics.

3



Original GP Benchmark Integrated

Original GP method successfully evaluated within the unified framework with automatic checkpointing.

4



Modified GP Benchmark Integrated

Modified GP variant evaluated using the same pipeline to enable a fair and controlled comparison.

5



EOH Search Successfully Executed

EOH framework executed an evolutionary search and obtained the best 8-feature extractor with high validation accuracy.

6



Common Evaluation Pipeline Established

All methods now use identical data splits, classifier and evaluation metrics, ensuring controlled comparisons.



WEEKLY HIGHLIGHTS



End-to-end benchmarking framework is now fully operational.



First controlled EOH result obtained: **92.86%** validation accuracy with 8 features.



All methods produce feature vectors with the same dimensionality for fair comparison.



Automatic logging, checkpointing and result summaries implemented.



The project has moved from implementation to rigorous experimental comparison.



KEY MESSAGE

We have now built a complete and reproducible experimental framework, obtained the first strong EOH result, and integrated all core methods. This positions us to perform a rigorous comparison of explainable feature extraction techniques.

Unified Benchmark Framework



We have developed a single, reproducible pipeline that ensures all feature extraction methods are evaluated under identical conditions, enabling a fair and controlled comparison.

FRAMEWORK COMPONENTS



Common Preprocessing

All images resized, normalized and converted to grayscale.



Common Data Splits

Identical train (60%), validation (20%) and test (20%) splits for every method.



Shared Classifier

Linear SVM with the same hyperparameters for all methods.



Common Evaluation Metrics

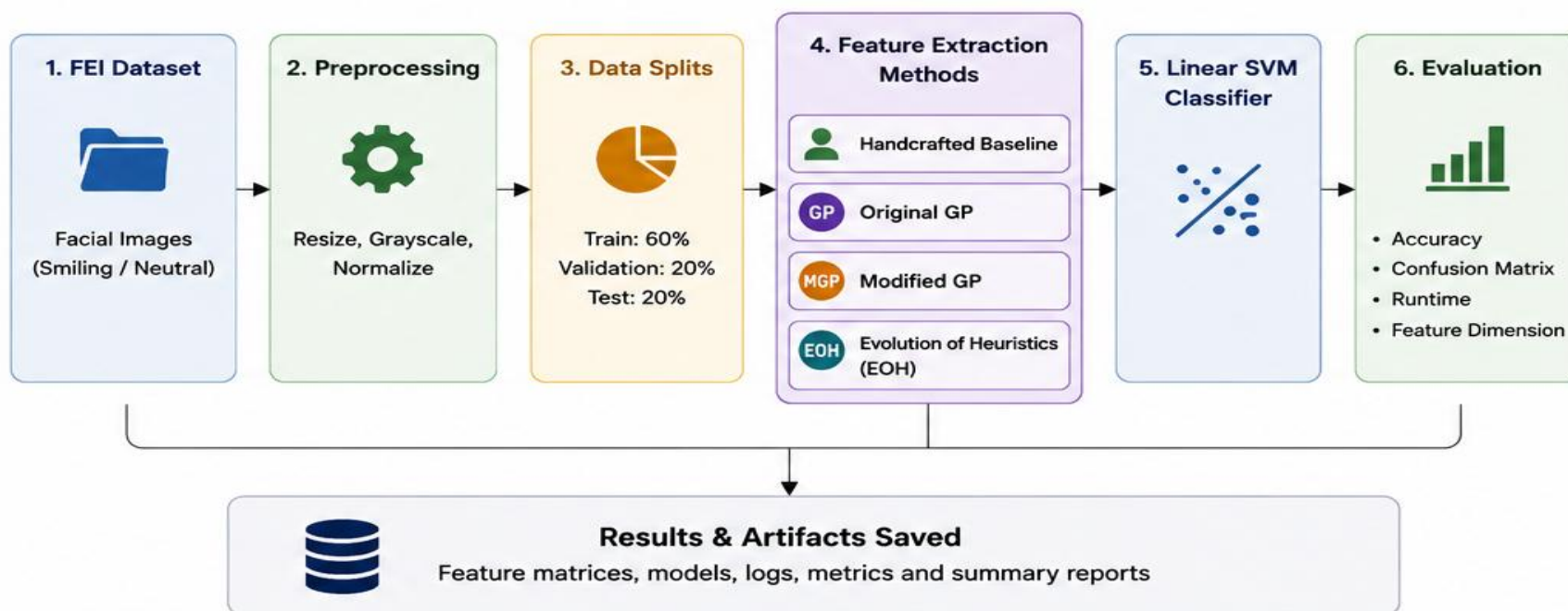
Accuracy, confusion matrix, runtime and feature dimension.



Automatic Logging & Checkpointing

All results, models and feature matrices saved automatically.

BENCHMARKING PIPELINE



KEY MESSAGE

This unified framework ensures that any differences in performance are due to the feature extraction method itself—not differences in data handling, preprocessing, classifier choice or evaluation.

Feature Dimension

8

(Fixed across all methods)

Benchmark Notebook Progress



The unified benchmark notebook now provides a **fully reproducible** and **automated evaluation environment** for all feature extraction methods.

FRAMEWORK CAPABILITIES



Common Preprocessing

Images resized, converted to grayscale and normalised in a consistent way for all methods.



Common Data Splits

Identical train (60%), validation (20%) and test (20%) splits used across all methods.



Shared Linear SVM Classifier

Same hyperparameters and training procedure ensuring a fair comparison.



Standardised Evaluation Metrics

Accuracy, confusion matrix, runtime and feature dimension recorded for every method.



Automatic Logging & Checkpointing

All models, feature matrices and logs saved automatically for reproducibility.



Benchmark Summary Reports

Automatic generation of summary tables and plots for easy comparison.

EXPERIMENTAL SETUP

Dataset	FEI Facial Expression Images
Total Images	700
Subjects	50
Images per Subject	14
Image Size	64 × 64 (grayscale)
Train / Val / Test Split	60% / 20% / 20%
Feature Dimension	8 (fixed across all methods)
Classifier	Linear SVM
Evaluation Metrics	Accuracy, Confusion Matrix, Runtime, Feature Dimension



KEY BENEFIT

All methods are evaluated under identical conditions, allowing performance differences to be attributed solely to the feature extraction strategy.

INTEGRATED METHODS



Handcrafted Baseline

Status: Integrated
8 interpretable features.



Original GP

Status: Integrated
Baseline GP implementation.



Modified GP

Status: Integrated
Modified operators and selection.



Evolution of Heuristics (EOH)

Status: Integrated
LLM-guided feature discovery.



Compact GP

Status: In Progress
Dimensionality control in development.



KEY MESSAGE

We now have a complete and reproducible benchmarking framework with four methods fully integrated. This ensures a robust and fair comparison of explainable feature extraction techniques.

Handcrafted Baseline

Reference Method and Control Experiment



An **explainable baseline** using **eight interpretable features** provides the reference point for evaluating all automated feature extraction methods.

8 INTERPRETABLE FEATURES

1		ROI Mean Intensity	Average pixel intensity within the facial region.
2		ROI Standard Deviation	Pixel intensity variability within the facial region.
3		Frame Difference Mean	Mean absolute difference between consecutive frames.
4		Frame Difference Std	Standard deviation of frame differences.
5		Bright Pixel Count	Number of pixels above a fixed intensity threshold.
6		Contrast (RMS)	Root mean square contrast of the image.
7		Skewness	Skewness of pixel intensity distribution.
8		Kurtosis	Kurtosis of pixel intensity distribution.

BASELINE PERFORMANCE (VALIDATION SET)

 Validation Accuracy 71.43% (50 / 70 correct)	 Feature Dimension 8 (Fixed)	 Feature Extraction Time 0.055 s (Per run)	 Classifier Training Time 0.006 s (Per run)
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EXAMPLE: FEATURE VECTOR (ONE IMAGE)

f_1	f_2	f_3	f_4	f_5	f_6	f_7	f_8
0.612	0.128	0.243	0.105	153.0	0.487	-0.214	2.135

WHY THIS BASELINE IS IMPORTANT

- ✓ Fully explainable and interpretable features with clear semantic meaning.
- ✓ Provides a strong control experiment for assessing automated methods.
- ✓ Ensures that performance improvements are due to better feature discovery, not differences in evaluation conditions.



KEY TAKEAWAY

The handcrafted baseline achieves 71.43% validation accuracy using only eight interpretable features. It serves as the explainable reference against which all GP and EOH methods are compared.

Evolution of Heuristics (EOH)

First Successful Evolutionary Search

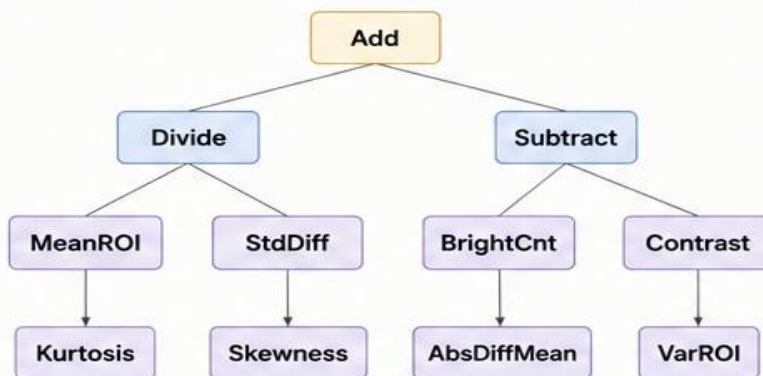


EOH was successfully executed using the DeepSeek LLM to evolve an interpretable feature extractor. The best individual discovered achieved **92.86% validation accuracy** with only **eight features**.

KEY RESULTS (BEST INDIVIDUAL)

 Validation Accuracy	92.86% (65 / 70 correct)
 Feature Dimension	8 (Fixed)
 Fitness (Error Rate)	0.071429 (1 – Accuracy)
 Feature Extraction Time	0.022 s (Per run)
 Search Method	Evolution of Heuristics (LLM-guided)
 LLM Used	DeepSeek (via API)
 Generations	20
 Population Size	20
 Early Stopping	Yes (No improvement for 5 gens)

EVOLVED FEATURE EXTRACTOR (BEST INDIVIDUAL)



Feature Descriptions

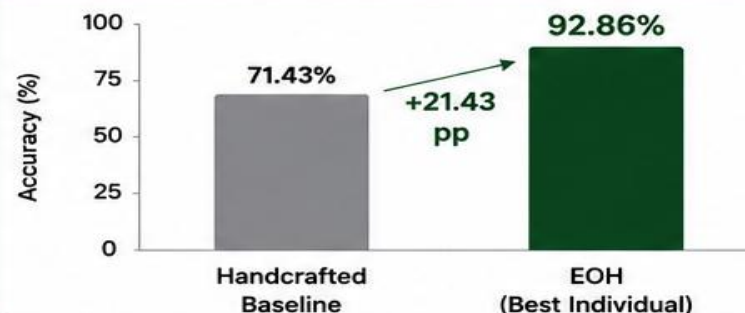
1. **MeanROI** – Mean intensity in facial ROI
2. **StdDiff** – Std. deviation of frame differences
3. **BrightCnt** – Count of bright pixels
4. **Contrast** – RMS contrast of the image
5. **Kurtosis** – Kurtosis of pixel intensities
6. **Skewness** – Skewness of pixel intensities
7. **AbsDiffMean** – Mean absolute frame difference
8. **VarROI** – Variance of intensity in ROI



KEY ACHIEVEMENTS

- First successful evolutionary search using LLM guidance.
- Discovered a high-performing, fully interpretable 8-feature extractor.
- Significant improvement over handcrafted baseline (71.43% → 92.86%).
- Satisfies the dimensionality constraint for fair comparison.

ACCURACY COMPARISON (VALIDATION SET)



KEY MESSAGE

EOH has delivered the first strong result of the project, achieving 92.86% validation accuracy with only eight interpretable features through automatic discovery guided by DeepSeek.



21.43 percentage point improvement over the handcrafted baseline.

10 What We Plan to Do Next

Focus: Compact GP and Robustness



Our next phase focuses on **creating truly compact GP feature extractors** and systematically **evaluating their robustness**, ensuring they generalise well beyond the data and conditions they were evolved on.



1. DEVELOPING COMPACT GP EXTRACTORS

We will refine our GP search to prioritise **simplicity and efficiency** while maintaining high predictive performance and interpretability.



Stronger Complexity Control

Tighten tree size and depth limits and apply stronger complexity penalties in the fitness function.



Parsimony Pressure

Explicitly encourage shorter, more interpretable trees using parsimony-aware techniques.



Operator Tuning

Adjust crossover and mutation settings to favour compact building blocks and reduce bloat.



Retain Interpretability

Ensure resulting trees use a small set of meaningful operations to produce clear, human-understandable features.



Outcome: A set of high-performing, compact GP feature extractors that are easy to understand and efficient to compute.



2. INVESTIGATING ROBUSTNESS

We will evaluate how well our evolved features **generalise** across different conditions, data variations and potential real-world challenges.



Data Perturbations

Test performance under noise, blur, brightness/contrast changes, scaling and other image-level perturbations.



Subject and Domain Variation

Evaluate across different subjects and unseen datasets to assess cross-subject and cross-domain generalisation.



Temporal Robustness

Check stability of performance across different subsets of frames and under temporal shifts.



Overfitting Assessment

Compare train/validation/test performance gaps and examine sensitivity to hyperparameters.



Statistical Reliability

Use appropriate statistical tests and confidence intervals to confirm that improvements are robust and significant.



Outcome: Confidence that our compact GP features generalise well and remain reliable under a wide range of conditions.



KEY TAKEAWAY

We will focus on building compact, interpretable GP feature extractors and ensuring they are robust, stable and generalise well — enabling reliable performance in real-world scenarios.



Goal: Compact, interpretable and robust feature extractors that generalise.